

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-01

BOTANY

ARSHID BHAT SIR



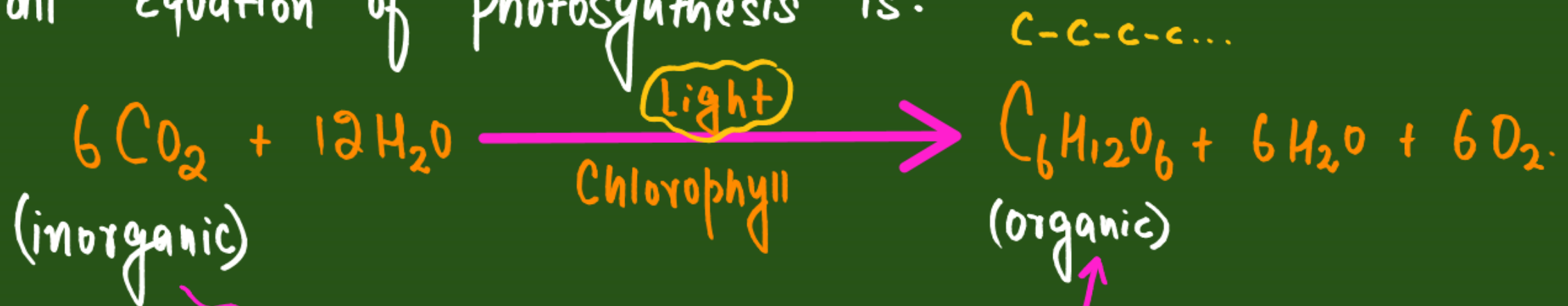
Topics To Covered

- 1** Introduction
- 2** Experiments Related to Photosynthesis
- 3** ...

INTRODUCTION



≡ Overall Equation of Photosynthesis is:



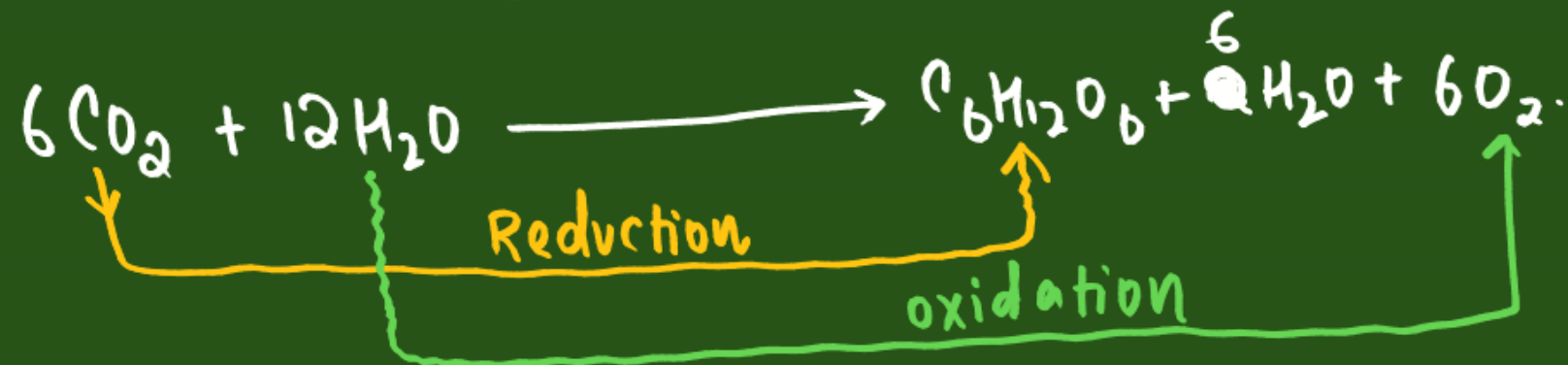
≡ Photosynthesis is a process in which organic glucose is formed from inorganic CO_2 .

≡ In photosynthesis $\left\{ \begin{array}{l} \text{CO}_2 \text{ is converted into glucose} \\ \text{light energy is converted into} \\ \text{chemical energy (C}_6\text{H}_{12}\text{O}_6\text{)} \end{array} \right\} \therefore \text{Photosynthesis is a physico-chemical process.}$

≡ Photosynthesis is Endothermic / Endergonic process — because Energy is used in this process.

≡ $C_6H_{12}O_6$ (Glucose) is formed from CO_2 (6 molecules) → i.e, why photosynthesis is Anabolic (Build-up) process.

≡ Photosynthesis is **Redox Reaction** (Reduction + Oxidation).



⊗ CO_2 gets reduced to form $C_6H_{12}O_6$. (Reduction)

⊗ H_2O gets oxidised to form O_2 . (Oxidation).

- Photosynthesis  Multistep process.
Enzymatically controlled process.
- Photosynthesis — is a process by which Chlorophyll containing organism prepare their food. (Autotrophs).

PHOTOSYNTHESIS

1. Oxygenic photosynthesis

- O_2 is by-product
- H-donor is H_2O .



Ex. → All Chlorophyll containing Eukaryotic organisms.

→ Cyanobacteria (BGA).

2. Anoxygenic photosynthesis

- By-product is other than O_2 (Ex. Sulfur).
- H-donor is other than H_2O (Ex. H_2S)

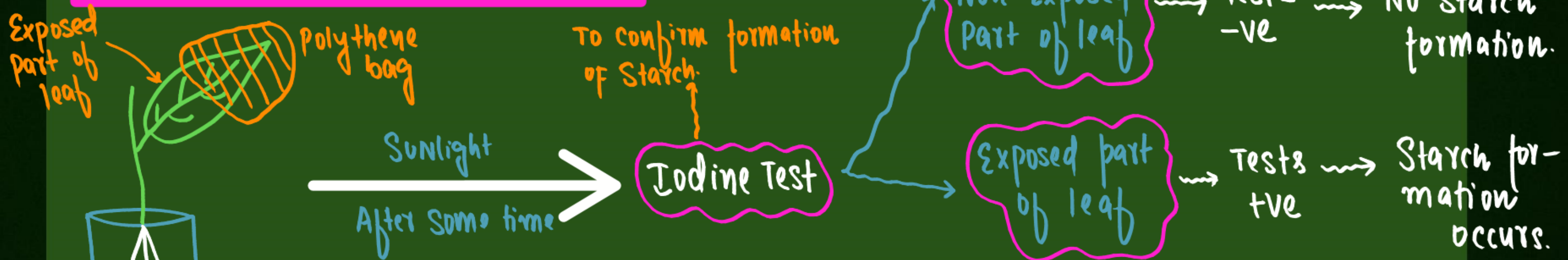


Ex. Purple Sulfur Bacteria.

Experiments Related to Photosynthesis

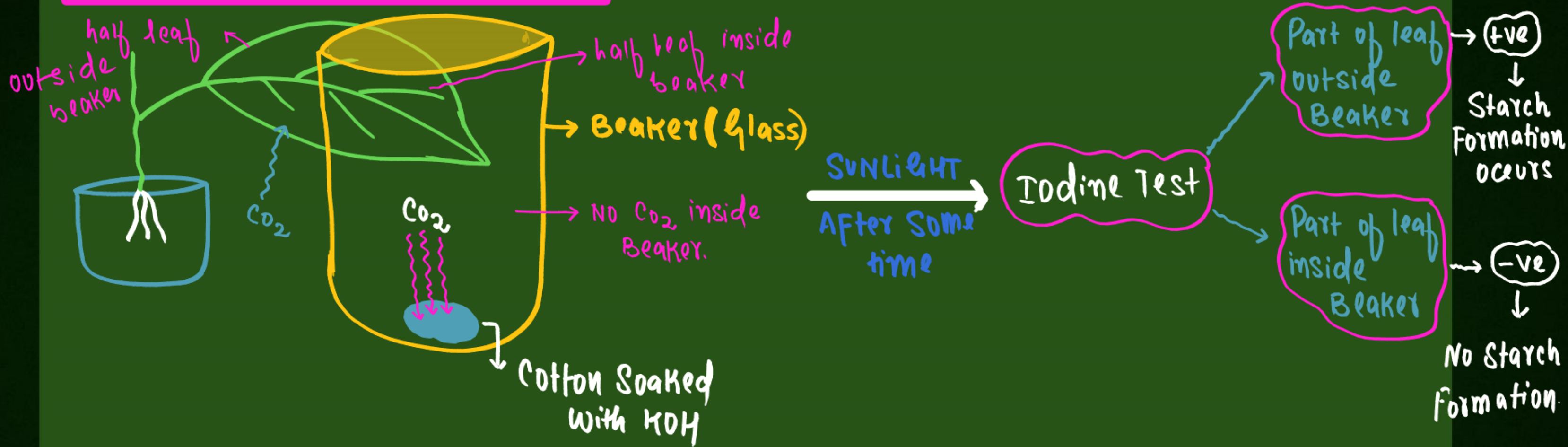
OLD Experiments

1. Variegated leaf experiment:



Conclusion Light is required for photosynthesis.

2. Mohl's half leaf experiment: (VON MOHL)



CONCLUSION

CO_2 is required for photosynthesis.

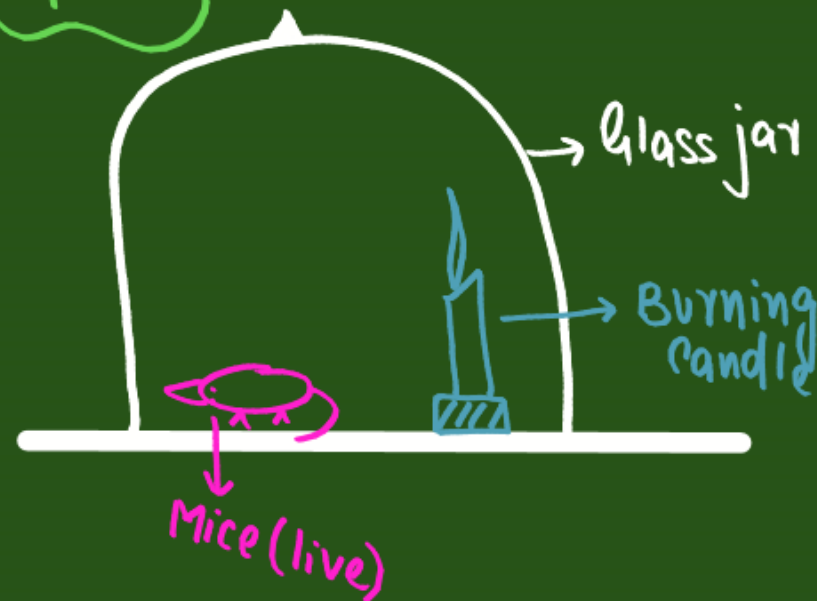
EARLY EXPERIMENTS

1) Bell jar experiment:

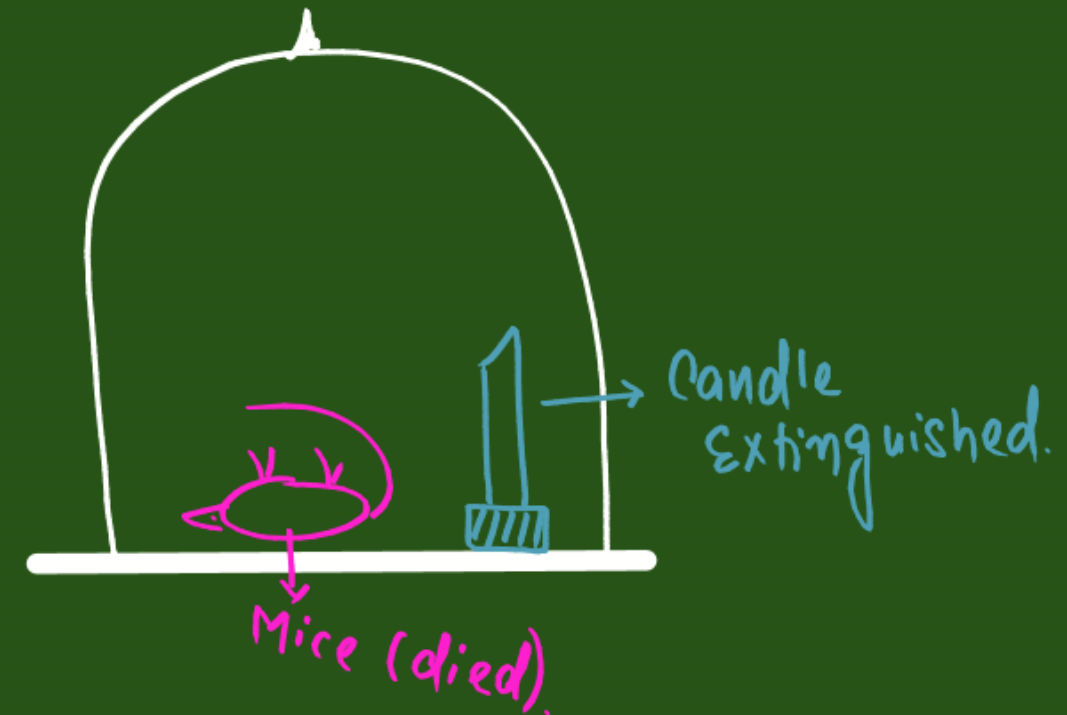
- performed by Joseph Priestly (1770).
- He also discovered Oxygen in 1774.

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He performed set of experiments:

SETUP 1



After Some time



CONCLUSION

Burning candle and living mice breathe the air and somehow damaged the quality of air.

Set-up 2

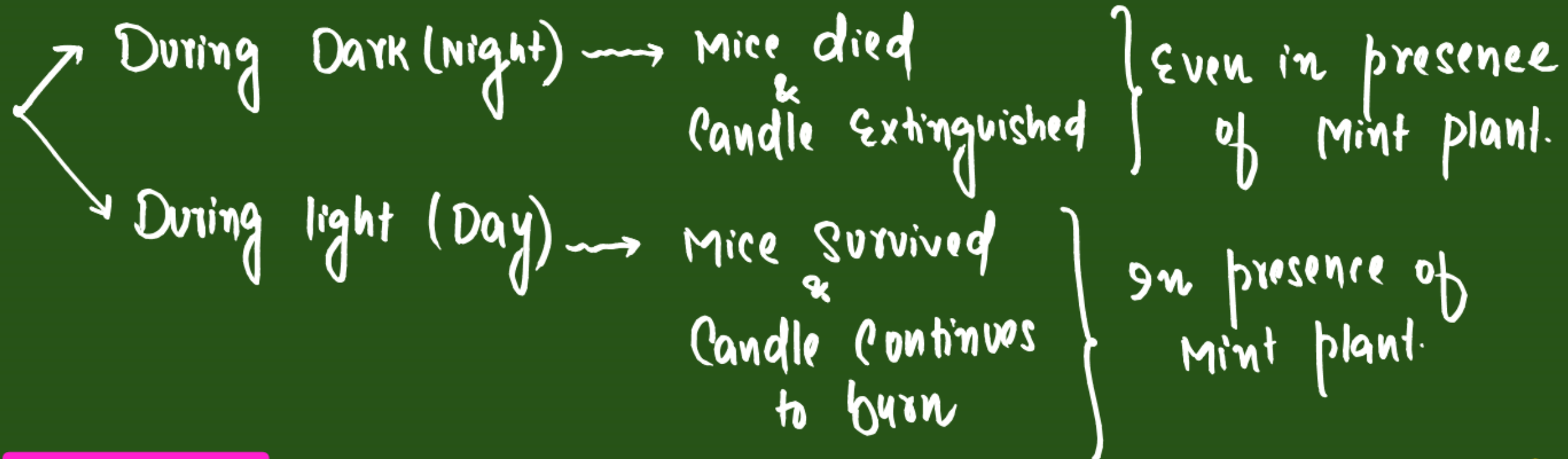


CONCLUSION

Plant (ex. Mint plants) restores to the air whatever breathing mice and burning candle removed.

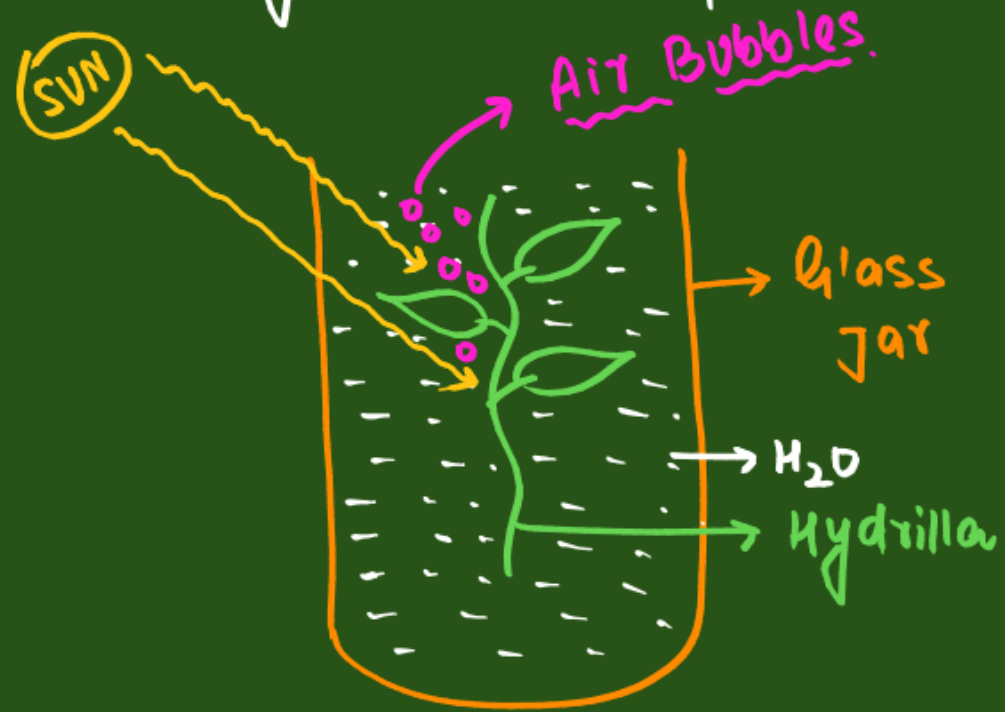
2) Jan Ingenhousz: (1730–1799)

- He performed Joseph Priestly's Experiment (Setup-2)
 - once in Dark. (Night time).
 - once in light. (Daytime)



CONCLUSION Sunlight is essential for the plant process that somehow purifies the air damaged by breathing mice and burning candle.

NOTE: Ingenhousz — performed another experiment by using hydrilla (Aquatic plant).



- He observed formation of air bubbles from green parts of plant which were exposed to sunlight.
- He later identified these air bubbles as oxygen.

3) T.W Engelmann: (1843–1909)

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Next lecture

4) Julius Von Sachs: (1854)

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Next lecture

5) Cornelius Von Neil: (1897–1985)

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Next lecture

6) Ruben and Kamen:

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QUESTIONS

“CO₂ is required for photosynthesis”. This conclusion can be drawn by performing

- A** Variegated leaf experiment
- B** Mohl's half leaf experiment
- C** Bell-jar experiment
- D** C. Niel experiment

HOMework

→ Revision
→ Don't Attempt DPP.



Thank
You